

Vendor Invoice Assistant

An AI-powered vendor self-service capability delivered over WhatsApp — moving routine invoice and payment enquiries off the Accounts Payable desk while preserving a clean, controlled path to a human officer.

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SECTION 01 · EXECUTIVE SUMMARY

From a manual enquiry desk to AI-mediated vendor self-service.

The Accounts Payable team today acts as a human-mediated enquiry desk for the entire vendor base. The proposed Vendor Invoice Assistant deflects the routine majority of those enquiries to an always-available conversational agent on WhatsApp, with a clean path to a human AP officer whenever it is needed.

~70%

Routine enquiries resolved without human touch

24x7

Vendor self-service availability

<10s

Mean first response time

1channel

WhatsApp — already adopted by vendors

EXECUTIVE POSITION

The Vendor Invoice Assistant converts a recurring operational cost — manually answering invoice-status questions — into a scalable digital capability. It runs on Azure under existing enterprise controls, uses a channel vendors already trust, and never modifies the AP system of record.

SECTION 02 · BUSINESS CONTEXT

Accounts Payable shapes vendor trust and working capital.

AP is the operational interface to the supplier base. Its responsiveness directly affects vendor confidence and the negotiating position of procurement.

VOLUME

Invoice and active-vendor volumes grow faster than the AP headcount available to answer enquiries.

VARIABILITY

Enquiry spikes around month-end and payment runs routinely exceed steady-state capacity.

VISIBILITY

Vendors have no read-side window into approval status, blockers, or planned payment dates.

Why this matters now

- WhatsApp Business Platform has matured into a compliant enterprise channel.
- Azure OpenAI provides governed, region-resident access to frontier models with tool-calling.
- Vendor expectations around responsiveness have moved firmly to "self-service in seconds".

SECTION 03 · CURRENT CHALLENGES

Where the existing model breaks down.

OPERATIONAL

AP officers spend the majority of their day on low-complexity, repeat status enquiries instead of exceptions.

EXPERIENCE

Vendors describe the process as opaque, slow, and inconsistent across regions and officers.

GOVERNANCE

There is no structured, queryable record of enquiries — disputes are reconstructed from email threads.

CHALLENGE	TODAY	OPERATIONAL IMPACT
Manual AP workload	No self-service surface for vendors	Capacity consumed by repetitive lookups
Delayed status communication	Synchronous human-only responses	1–3 business day turnaround
Inconsistent answers	Multiple officers, multiple sources	Knowledge lives in people, not systems
Lack of self-service	Existing vendor portal is under-adopted	Email becomes the default queue
No audit trail	Conversations happen in email and phone	Disputes hard to defend
Limited working hours	Single-region human team	Global vendors poorly served

SECTION 04 · CURRENT VS FUTURE STATE

The same job, a fundamentally different operating model.

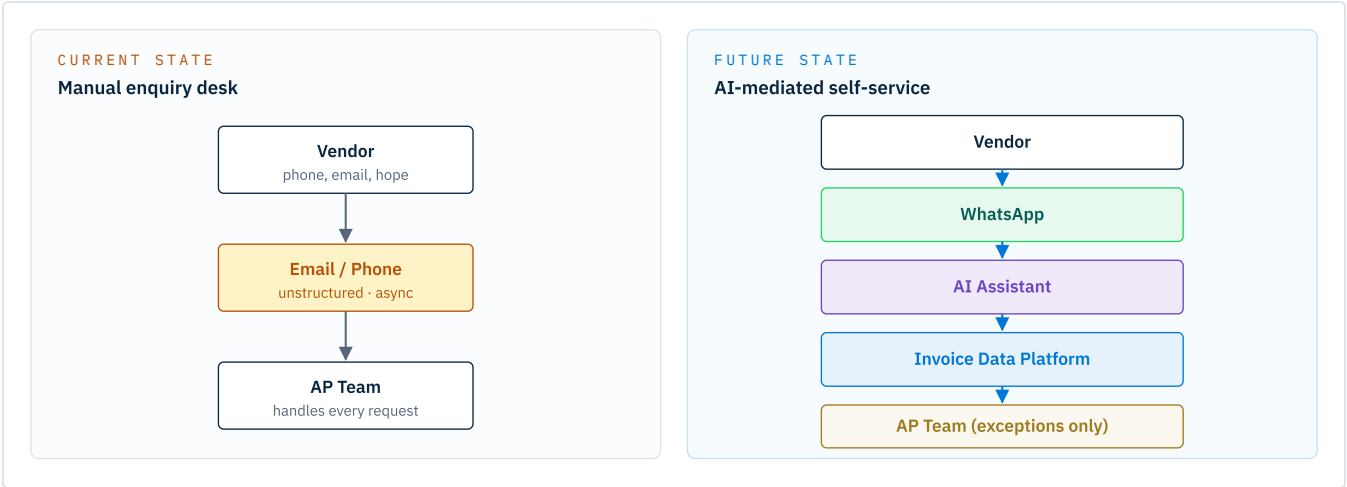


FIG 4.1 THE SAME BUSINESS JOB, EXECUTED BY A DIFFERENT OPERATING MODEL.

Measurable improvements

METRIC	CURRENT	FUTURE (YEAR 1)	IMPROVEMENT
Mean vendor response time	~ 6 hours	< 10 seconds	~ 2,000× faster
Self-service deflection	0%	≥ 70%	+70 pts
AP enquiry volume (email)	Index 100	≤ Index 50	–50%
Availability	Business hours, one region	24×7 global	3×
Audit trail of enquiries	Partial (email)	Structured, queryable	Full
AP team focus	Status lookups	Exceptions & disputes	Higher-value work

SECTION 05 • PROPOSED SOLUTION

An AI-orchestrated assistant on WhatsApp, governed by Azure.

A conversational, AI-mediated layer that sits between vendors (on WhatsApp) and the Invoice Data Platform. Azure OpenAI interprets intent, retrieves only the vendor's own data, generates the reply, and decides when to escalate.

DESIGN PRINCIPLES

- **AI-first.** The model orchestrates the conversation — not a hand-built menu tree.
- **Channel-native.** WhatsApp is treated as a first-class surface with lists, buttons, and confirmations.
- **Implicit identity.** Vendors are identified by their registered WhatsApp number; no login.
- **System of record stays canonical.** The assistant reads from the AP datastore; the ERP remains the source.
- **Human-in-the-loop.** Escalation is a first-class flow, not a failure mode.

WHAT IT DOES, IN ONE SCREEN

- Confirms whether an invoice has been received and the stage it is in.
- Shares expected and actual payment dates.
- Explains rejection reasons and next steps.
- Lists outstanding invoices and their status.
- Captures disputes as structured records.
- Escalates cleanly to the AP team with full conversation context.

WHY NOW

The combination of an enterprise-ready conversational channel (WhatsApp Business Platform) and governed, region-pinned frontier models (Azure OpenAI) turns what would previously have been a bespoke build into a configurable delivery on existing enterprise infrastructure.

SECTION 06 · BUSINESS BENEFITS

What the organisation gets — and how it is measured.

✓ FASTER VENDOR SUPPORT

Conversational answers in seconds, not days. 24×7 across time zones.

✓ REDUCED AP WORKLOAD

Routine enquiries deflected; team focuses on exceptions and disputes.

✓ BETTER VENDOR EXPERIENCE

Native channel, no portal login, consistent answers grounded in the system of record.

✓ AUDITABILITY

Every interaction logged as structured conversation history — disputes have a clear trail.

✓ WORKING-CAPITAL IMPACT

Faster identification of rejections and blockers accelerates payment cycles.

✓ PLATFORM FOUNDATION

Becomes the basis for further finance-ops AI services under the same controls.

Year-1 targets

METRIC	BASELINE	TARGET	YEAR-1 INDICATIVE VALUE
Deflection rate	0%	≥ 70%	1.5–2.5 FTE released
First-response time	~ 6 h	< 10 s	~ 2,000× faster
AP mailbox volume	Index 100	≤ Index 50	–50%
Vendor CSAT	Not measured	≥ 4.3 / 5	New baseline
Dispute capture completeness	~ 40%	≥ 95%	Structured records

SECTION 07 · HIGH-LEVEL ARCHITECTURE

One executive view of the end-to-end flow.

Three trust zones: the public internet, the WhatsApp platform, and the enterprise Azure estate. One orchestrator, two intelligence sources, one human escalation path.

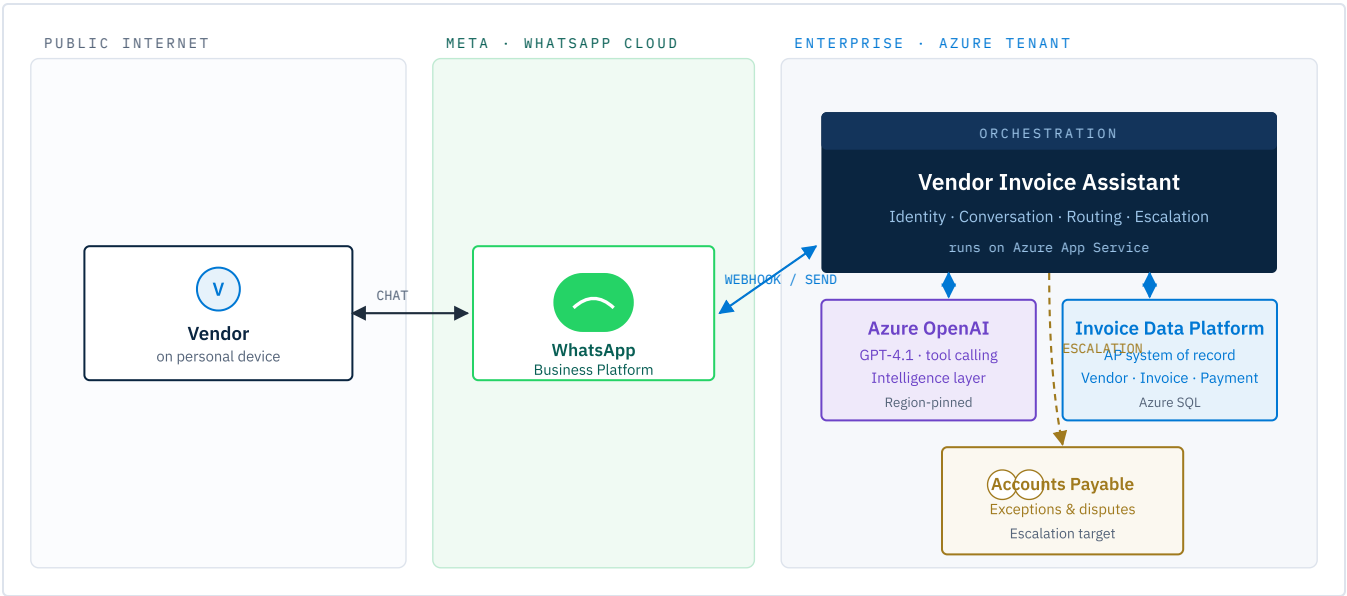


FIG 7.1 THREE TRUST ZONES, ONE ORCHESTRATOR, ONE ESCALATION PATH.

SECTION 08 · SOLUTION ARCHITECTURE

Six logical layers, six responsibilities.

A logical view of the platform. Each layer has a single responsibility and a defined contract with its neighbours. Implementation detail is deliberately kept out of this view.

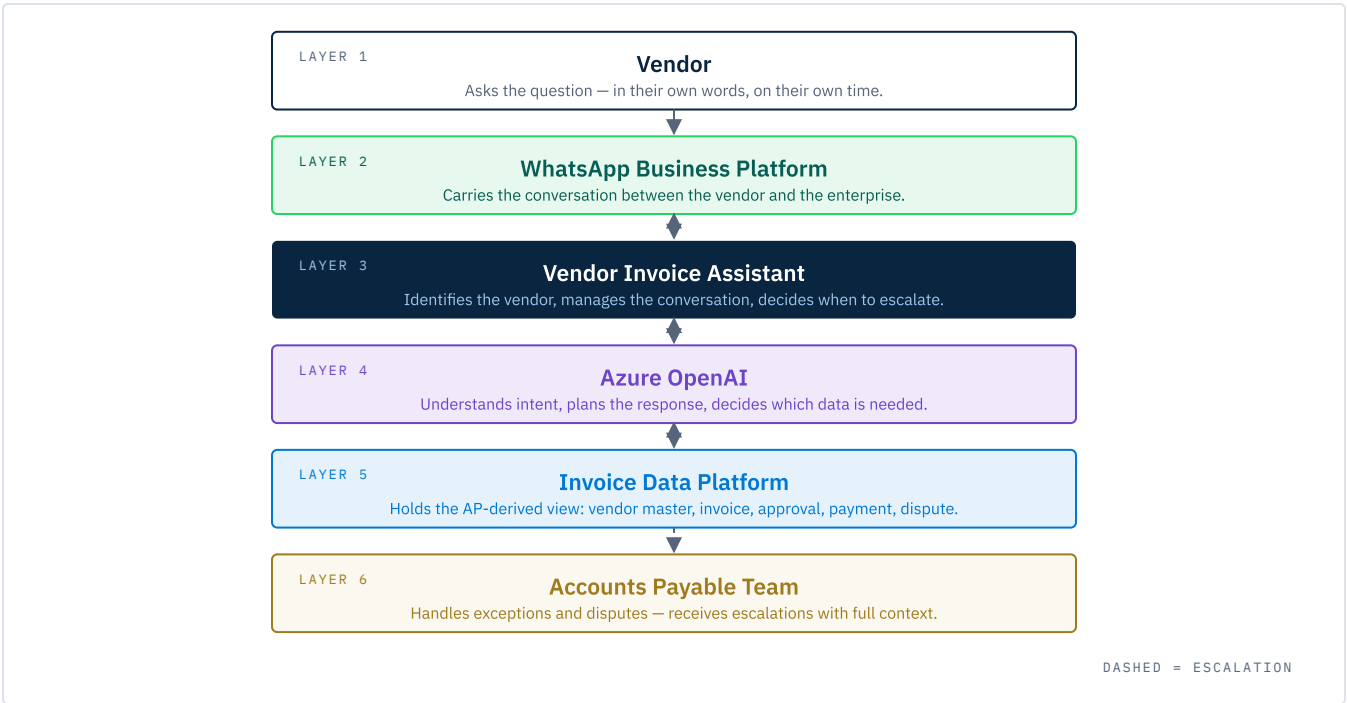


FIG 8.1 SIX-LAYER LOGICAL VIEW. EACH LAYER HAS ONE JOB.

SECTION 09 · AI ARCHITECTURE

An agentic loop, grounded in vendor-scoped data.

Azure OpenAI is the planner: given the vendor's message and context, it decides which data lookup to perform, evaluates the result, and either responds or escalates. The host application enforces what the model is allowed to do.

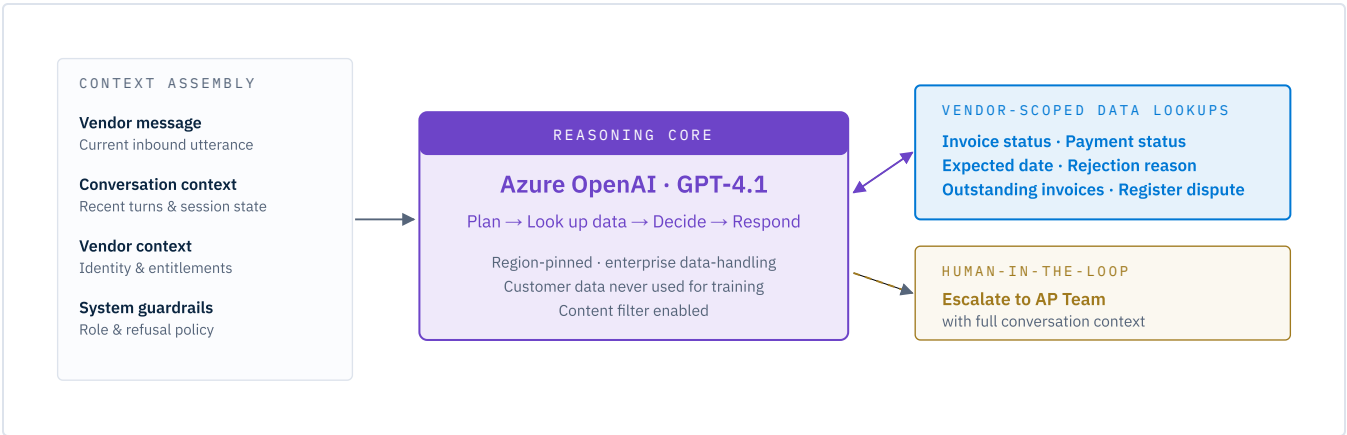


FIG 9.1 THE MODEL IS THE PLANNER; TOOLS ARE VENDOR-SCOPED; HUMANS ARE THE SAFETY NET.

CORE PROPERTY – VENDOR ISOLATION

The model never receives data from another vendor. Every data lookup resolves the vendor from the verified WhatsApp sender — not from any model-supplied argument — so the data boundary is enforced by construction.

SECTION 10 · WHATSAPP INTEGRATION ARCHITECTURE

How the platform binds to Meta Cloud API.

WhatsApp Business Platform is consumed via Meta Cloud API. A registered Meta Business owns a WhatsApp Business Account (WABA), which owns the dedicated business phone number. The Cloud API delivers inbound messages to a registered webhook and accepts outbound sends.

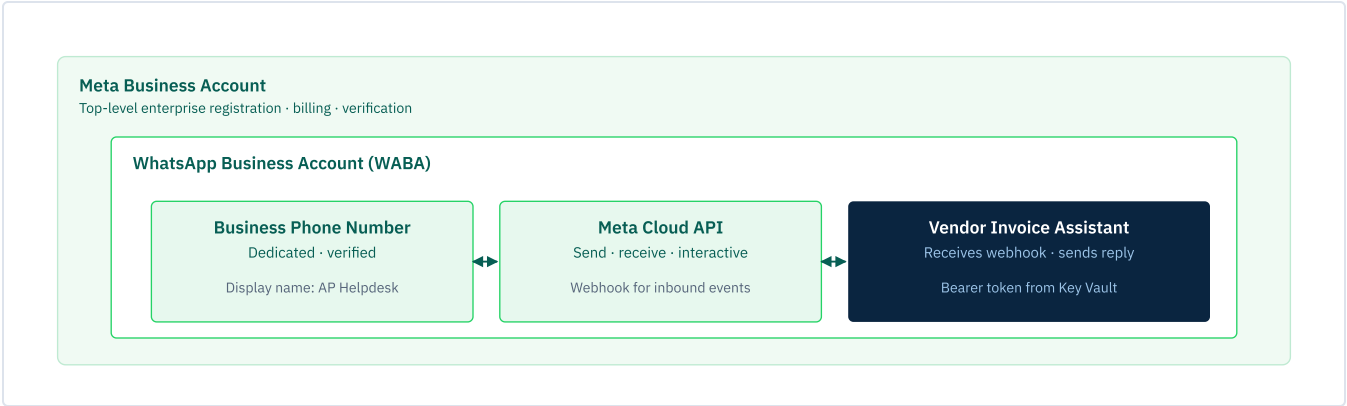


FIG 10.1 OWNERSHIP: META BUSINESS → WABA → NUMBER → CLOUD API → ASSISTANT.

Onboarding prerequisites

ITEM	OWNER
Meta Business Account & business verification	Finance ops
WhatsApp Business Account (WABA) created	Finance ops
Dedicated phone number provisioned & verified	Telephony
Display name approved by Meta	Brand & comms
Cloud API enabled & webhook registered	Platform engineering
Production access approved by Meta	Finance ops + engineering

OPERATIONAL NOTE

WhatsApp allows free-form replies only within the 24-hour customer-service window opened by a vendor message. Proactive notifications (e.g. "payment completed") require approved message templates — a Phase 4 capability.

SECTION 11 · DATA ARCHITECTURE

Six business data domains.

A business-level view of the data the assistant uses. The system of record stays in the ERP; the platform consumes a read-side view of it.

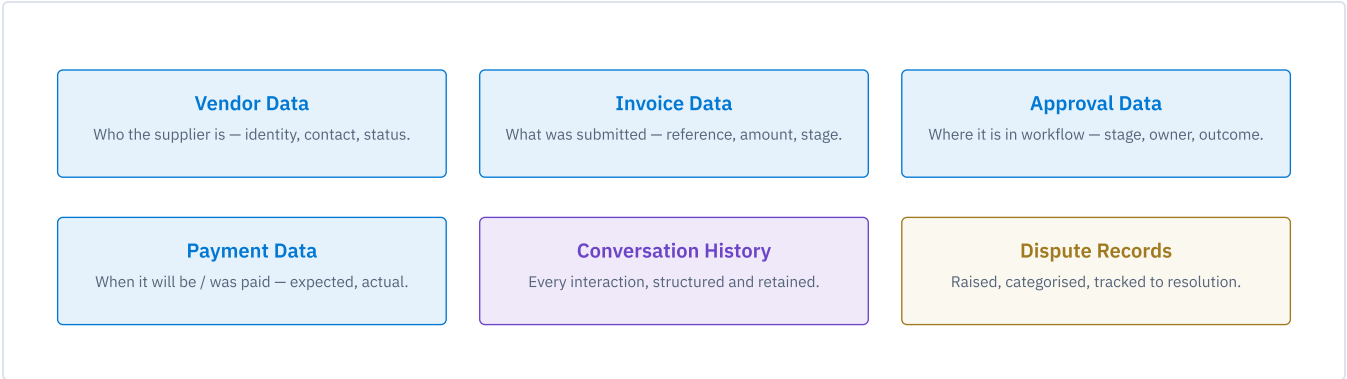


FIG 11.1 SIX LOGICAL DATA DOMAINS UNDERPIN THE ASSISTANT.

DATA RESIDENCY

All operational data resides within the elected Azure region. Conversation history and dispute records follow the enterprise retention policy for AP-adjacent records.

SECTION 12 • SECURITY ARCHITECTURE

Defence in depth with vendor-data boundary enforced server-side.

IDENTITY

Vendors are identified by their verified WhatsApp sender number, mapped to a vendor master record. Unknown senders are politely refused.

DATA SCOPING

Every data lookup filters strictly by the authenticated vendor — server-side, never from model input.

SECRET MANAGEMENT

WhatsApp tokens, model keys, and database credentials live in Azure Key Vault with rotation and audit.

TRANSPORT & INGRESS

TLS end-to-end; Azure Front Door + WAF in front of the webhook; signature verification on every inbound event.

AI GOVERNANCE

Region-pinned Azure OpenAI; customer data excluded from model training; content filter active; refusal policy for out-of-scope intents.

AUDITABILITY

Append-only log of every data lookup with actor, vendor scope, and outcome — non-repudiable record of what the assistant did.

COMPLIANCE POSTURE

Aligns with enterprise data-protection policy. Right-to-erasure for vendor conversations supported through retention controls on the Conversation History domain. Microsoft enterprise data-handling commitments apply to Azure OpenAI usage.

SECTION 13 · AZURE DEPLOYMENT ARCHITECTURE

Production topology on Microsoft Azure.

PaaS services within a single subscription per environment (Dev / Pilot / Production), wired through managed identity, secrets in Key Vault, observability through Application Insights.

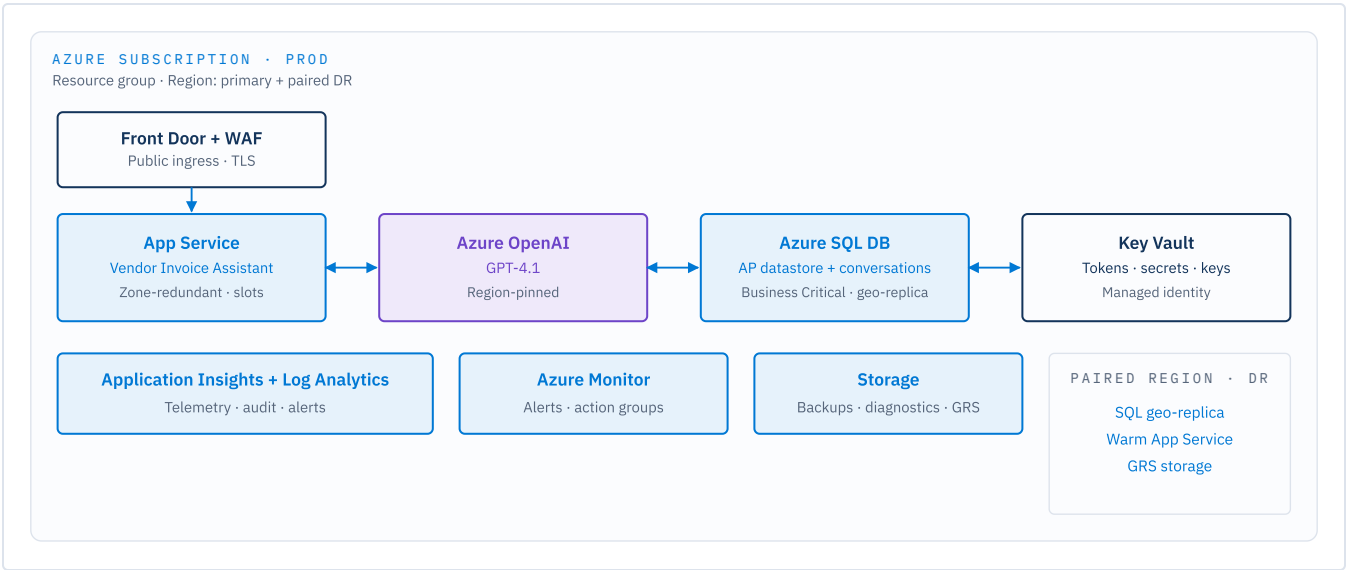


FIG 13.1 AZURE TOPOLOGY — MANAGED PAAS, GEO-PAIRED DR.

SECTION 14 • MONITORING

Four monitoring planes, four executive questions.

APPLICATION MONITORING

Is the platform up and responsive?
Request volume, latency, error rate,
dependency health.

AI MONITORING

Is the model behaving well? Tokens
consumed, model latency, refusal rate,
escalation rate, drift.

SECURITY MONITORING

Is anyone trying to do something they
shouldn't? Signature failures, unknown
senders, scope-violation attempts.

BUSINESS KPI MONITORING

Is the platform delivering value? Deflection
rate, response time, CSAT, top intents,
dispute funnel.

THREE ROLE-BASED DASHBOARDS

SRE Health — latency, errors, dependencies. **AI Governance** — tokens, refusal, escalation, drift.
AP Leadership — deflection, CSAT, top intents, dispute funnel.

SECTION 15 · COST ESTIMATION

Indicative Azure run-cost across POC, Pilot, and Production.

Figures are indicative, in USD, at Azure list pricing in a representative commercial region. Excludes one-time onboarding (Meta verification, integration), labour, and licensing. Final figures depend on enterprise discounts and chosen region.

\$1.6k /mo

POC · ~\$53 / day · ~\$19k / year

\$5.2k /mo

Pilot · ~\$173 / day · ~\$62k / year

\$14k /mo

Production · ~\$467 / day · ~\$168k / year

Cost by phase

PHASE	DAILY (USD)	MONTHLY (USD)	YEARLY (USD)	SCOPE
POC	~ 53	~ 1,600	~ 19,200	5 internal users · 1 month
Pilot	~ 173	~ 5,200	~ 62,400	~100 vendors · 3 months
Production	~ 467	~ 14,000	~ 168,000	Full vendor base · steady-state

WhatsApp Assistant vs Vendor Portal — comparison

DIMENSION	WHATSAPP ASSISTANT (PROPOSED)	VENDOR PORTAL (ALTERNATIVE)
Build cost (indicative)	\$120k–180k	\$350k–500k
Monthly run cost (production)	~ \$14k	~ \$11k
Vendor adoption barrier	None — uses channel vendors already have	High — portal login, password reset, training
Time to first value	~ 2 months (POC)	~ 6–9 months (build & rollout)
24×7 self-service	Yes, native	Yes, if vendor logs in
Conversational interface	Yes, AI-mediated	No — form-based
Expected deflection rate	~ 70%	~ 30% (industry average)
Mobile-first	Yes, by design	Requires responsive build
Vendor experience	Conversational, seconds	Transactional, multi-click
Strategic optionality	Reusable AI platform for other finance ops	Single-purpose web app

RECOMMENDATION

Proceed with the WhatsApp Assistant. Lower build cost, materially shorter time-to-value, dramatically higher expected adoption, and the resulting AI platform is reusable across finance operations. The marginal monthly run-cost difference vs a portal is recovered many times over by the deflection gain.

SECTION 16 · PRODUCTION READINESS

What "ready for production" means here.

DIMENSION	TARGET POSTURE	STATUS
Availability	≥ 99.9% monthly for the vendor-facing path	On track
Security	No vendor-data exposure incident	On track
Scalability	5× steady-state spike absorbed without manual action	On track
Disaster recovery	RPO ≤ 5 min · RTO ≤ 60 min	Pilot deliverable
Observability	Every request traceable; role-based dashboards	On track
Operational support	24×7 on-call with documented runbooks	Pilot deliverable
Business continuity	Graceful degradation to AP-only when AI is down	Pilot deliverable

PRE-GO-LIVE ACTIONS

(1) DR game-day with regional failover. (2) Approved runbooks for "AI down" and "WhatsApp down" graceful-degradation. (3) AI governance sign-off on refusal-policy thresholds. (4) Capacity-reserved OpenAI tokens for peak windows.

SECTION 17 · FUTURE ROADMAP

From POC to enterprise platform — four phases.

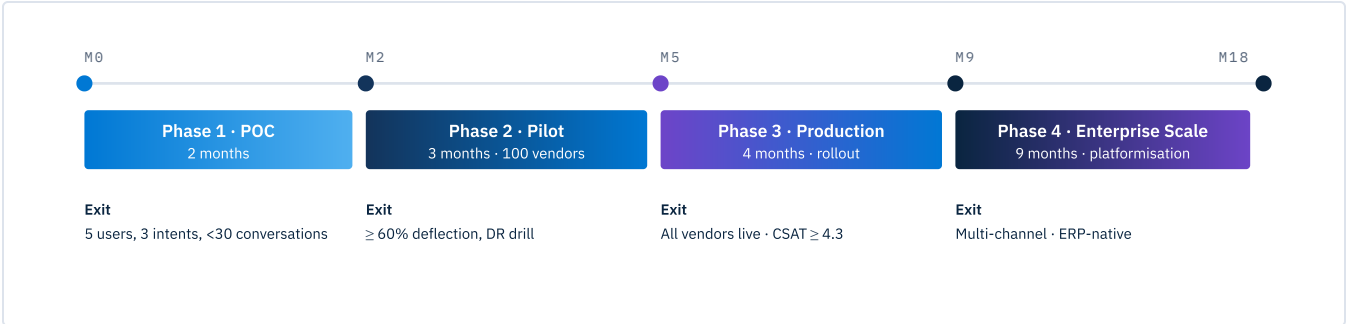


FIG 17.1 FOUR PHASES WITH EXPLICIT EXIT CRITERIA AND FUNDING GATES.

Phase 4 enhancement options

ERP INTEGRATION

Move to native ERP API calls for select intents.

EMAIL & VOICE CHANNELS

Same orchestrator, additional channel adapters.

MULTI-LANGUAGE

Localise to the languages of the vendor base.

PREDICTIVE INSIGHTS

"On-track / at-risk" proactive payment nudges.

VENDOR PORTAL COMPANION

Web surface for richer interactions and document upload.

ANALYTICS

Self-service ageing, payment trends, rejection patterns.

SECTION 18 • ARCHITECTURE DECISIONS

Four decisions that shape the platform.

DECISION 1 • AI-FIRST ARCHITECTURE

The AI model is the orchestrator, not a thin layer behind a rules engine. It interprets intent and decides which data to fetch. *Why*: dramatically richer handling and far lower long-term maintenance than hand-tuned intent trees.

DECISION 2 • WHATSAPP AS THE PRIMARY CHANNEL

Launch on WhatsApp. No portal, no email, no app first. *Why*: vendors already use it daily — adoption is effectively free. Email and portal added later, behind the same orchestrator.

DECISION 3 • HUMAN ESCALATION AS A FIRST-CLASS CAPABILITY

Escalation is built in as an explicit path the assistant or the vendor can invoke at any time. *Why*: trust and CSAT are demonstrably higher when an honest human path is visible.

DECISION 4 • AZURE OPENAI AS THE AI PLATFORM

Azure OpenAI over direct OpenAI APIs or self-hosted open-source. *Why*: region pinning, enterprise data-handling commitments, native managed-identity integration, and existing enterprise procurement governance.

SECTION 19 · EXECUTIVE RECOMMENDATION

Proceed with the POC.

RECOMMENDATION

Proceed to fund a **2-month POC** against three core intents (invoice status, payment status, rejection reason) with five internal AP users, on the Dev Azure subscription. POC exit is a green-lit funding gate for the 3-month, 100-vendor Pilot.

Why proceed

- **Operational return.** Targets 1.5–2.5 FTE released and a $\geq 50\%$ drop in AP mailbox volume within Year 1 of Production.
- **Vendor experience step-change.** Mean response time from ~6 hours to under 10 seconds.
- **Low POC risk.** Single subscription, three intents, five users, contained scope, ~ \$1.6k / month run-cost.
- **Strategic optionality.** The resulting AI platform is reusable across finance operations.

INDICATIVE ROI (YEAR 1, PRODUCTION)

Production run-cost	~ \$168k
FTE capacity released (~2 FTE · ~\$60k loaded)	~ \$120k
Working-capital benefit (faster resubmission, payment-cycle gains)	~ \$80k–150k
Vendor-experience & audit benefit	Qualitative
Net Year-1 position	Breakeven to + ~ \$100k

ROI compounds in Year 2+ as run-cost stays flat while deflection and reuse increase.

Key risks & mitigations

RISK	MITIGATION
Meta platform policy / pricing change	Architecture is channel-pluggable — email/voice/portal can be added behind the same orchestrator
Vendor-data exposure	Server-side vendor scoping by construction; append-only audit; refusal policy
Model behaviour drift	Region-pinned deployment, evaluation suite gates every prompt/model change, AI governance review
Low vendor adoption	Targeted onboarding with top-decile vendors first; KPI gate before broad rollout
Operational readiness gap at scale	Pilot exit criteria include DR drill and graceful-degradation runbooks

Recommended next steps

1. Approve POC funding and assign an executive sponsor (CFO or AP leadership).
2. Confirm Meta Business verification readiness and reserve a dedicated phone number.
3. Stand up the Dev Azure subscription with Azure OpenAI quota.
4. Identify the 5 internal pilot users and the 3 launch intents.
5. Set POC exit criteria and a Pilot funding-gate review date.

GO / NO-GO

GO — proceed with the 2-month POC. The combination of low POC cost, materially shorter time-to-value than the alternative, and the strategic optionality of an AI platform reusable across finance operations makes this a strong-conviction recommendation.